

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Integration	Connect Tableau to the state-wise consumption CSV/Excel source Refresh and load data into a Tableau extract (.hyper)
FR-2	Data Preparation	Clean and deduplicate records; standardize state names Handle missing values Create calculated fields: lockdown-period flag, YoY growth, recovery rate
FR-3	Data Visualization	Trend line chart of national/regional consumption (2019–2020) Regional bar chart and bubble map Calendar heat map of daily usage
FR-4	Dashboard	Interactive filters (state, region, date range) Filter actions linking multiple charts KPI summary cards
FR-5	Story	Guided narrative covering the three scenarios: overall trend shift, regional variation, lockdown recovery
FR-6	Publishing & Web Integration	Publish the workbook to Tableau Public / Tableau Server Embed the dashboard in a webpage for stakeholder access

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard and story are understandable by non-technical stakeholders within a few minutes of exploration, with no Tableau training required.
NFR-2	Performance	The dashboard and its filters respond within 2–3 seconds even across the full ~16,600-row dataset, using Tableau extracts rather than a live connection.
NFR-3	Accuracy	Calculated fields (YoY growth, recovery rate, lockdown flag) are validated against manual spot-checks before publishing.

NFR No.	Non-Functional Requirement	Description
NFR-4	Accessibility	The published dashboard is viewable on desktop and mobile browsers via Tableau Public/Server, without requiring a Tableau license.
NFR-5	Maintainability	The workbook is structured with clear worksheet and field naming so it can be updated as new data arrives.
NFR-6	Scalability	The extract-based design accommodates additional years of data or new states/regions without rebuilding the workbook from scratch.